

This is a briefing about my role at Integrated Solution for Ports (on-site job role), As one of the company team working in Adabia Port. We are a team content of 31 Persons

Our Mission is to ensure the system stability and guarantee a smooth workflow and deliver the fastest and reliable solutions among all cross-departments in the port, plus continue improvement to enhance the workflows and the system performance. our company is the provider of SPS the Port System. The Stack used for the system (Oracle DB, Java, Spring Boot, HTML, CSS, Apache Tomcat server and Microsoft Active MQ)

- engage with stakeholders at any Bug, change request or a custom report needs Query Script.
- Perform simple SQL queries over Toad Oracle, or using Crystal Report (developer version) to filter or edit reports.
- one of three persona act as whom to go to when anyone of the port staff operations or even the port customers, need help or have an issue with the system they can scale it up to me.
- handle the stages of the patches when it arrives from HQ, install it in test env define the UAT, test it and document it sends the review back if not valid else install it in the prod env. and provide the stakeholder with any explanation about it or a user guide if need it.
- prepare the documentation need it for of Bugs, CR's and custom reports from the backlog till the close state, I even built a custom app using frappe framework to watch out the request from the perspective of states, iterations and impacts on the system modules.
- responsible and leads about 10 employees in our team they work with the ports management department (customer support - yards and operations of loading and unloading department).
- providing the support for the terminals integrated with the system such as POS terminals used for payment at the check-out office for Trucks cashier and the terminals of Truck scales.
- contributing to define the issues in the integration messages because the system is integrated with MTS (Misr Technology Services- Nafaza) and MSW (Maritime Single Window). Mentioned

As a Self-starter and love my job I learned some tools and frameworks to help me organize my work, make it more easy, efficient and improve my competences. below are some of these tools

- frappe framework: which provide an enhanced and very customized env, I use it to build a custom app. and I touched the Maria DB through it.
- Power BI: for me one of the best tool to build semantic models and statistics report.
- HTML, CSS, JavaScript, React, JQuery, Bootstrap: which helped me to fully understand the web based application, as it helped me in the apps i build.
- Rest API's: while every application has routes, end points and server requests i have learned what's going behind the seen.
- SQL: every application has to interact with a database through a plan SQL query or ORM. it helped me a lot to understand how to elicit data, the relationships between and how the entities are linked.
- BPMN & Visio: to see the full picture of a workflow you cannot just imagine it you must see it in front and that what BPMN provided me with, it will make you pinpoint the workflow bottlenecks, optimize it and where you can implement the automations.
- GitHub, Linux: I cannot get involved in all what i mentioned above without theses tow.
- Analysis: Hybrid Research Methods, Root Cause Analysis, Protocol Analysis, User Role Analysis, Scenario Analysis, Stakeholder Management, Conflict Resolution, User Training and Problem Solving

these are my real weapons in my main role and as BABOK Guide mentioned A business analyst is any person who performs business analysis tasks no matter their job title or organizational role.

Selected Achievements which has a high influence in port community in particular in the system, each task was a challenge for me, many have been trying before but they end it with non-feasible.

Unify the Truck scales TCP Server:

- **Case:** there 11 truck scale points connected with the system through 3 different TCP servers using VSB and JAVA.
- **Problem:** 11 terminals / 3 Servers, mess in the documentation for each terminal uses which server which leads to hard to maintain and configuration and time consuming.
- **Solution:** By checking each terminal and log its out puts using Putty I was able to determine and categorize the common output among them creates one configuration file hold each terminal variables like start chart, end chart, Port number etc. inject that file to one Java TCP Server.
- **Results:** A single fact of truth for all terminals one TCP Server, comprehensive one user Guide as a reference for the Tech team to configuration the Terminals in 4x faster than before. plus, a possibility for scale or maintain in the feature.

Export Returns Workflow Optimization:

- **Case:** one editable record to register the export return.
- **Problem:** Identified a critical misalignment between port policy and system logic that led to data overwriting.
- **Solution:** Re-engineered the functional flow with automated quantity validation and versioning.
- **Results:** Eliminated manual entry errors and ensured 100% systematic policy enforcement

Unified Truck Impoundment Architecture:

- **Case:** 2 workflows to impoundment the truck for the check-In and the Check-Out.
- **Problem:** redundant workflows caused by rigid coupling between impoundments and Truck Visit Records.
- **Solution:** Facilitated cross-functional workshops to design a decoupled, lifecycle-based state management model for the trucks workflow.
- **Results:** Reduced operational complexity and provided a complete historical audit trail for legal compliance.

SDLC & Support Governance Application

- **Case:** Managing a high volume of system change requests (CRs) and bugs for the SPS system.
- **Problem:** Lack of visibility into request states, iterations, and their specific impacts on various system modules.
- **Solution:** Built a custom governance application using the **Frappe Framework** to track the entire lifecycle of system requests.
- **Results:** Streamlined the documentation process from backlog to closure and provided a real-time "impact map" for system modules, significantly reducing technical risk during deployments

Validating tool for Integration messages

- **Case:** High volume of errors in messages between the SPS and MTS (Nafaza).
- **Problem:** Complexity in identifying failure points within the integration message received from MTS, leading to potential delays in port clearance.
- **Solution:** Built a custom tool with frappe framework to parse and validate the content of the message provide the end user with the wrong content.
- **Results:** Reduced the time-to-resolution for integration errors, ensuring continuous compliance and smooth data flow.